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Open Work Rights · Post-Study Work Visa

Yi Xia *(Sean)*

AI Engineer · Computer Vision Researcher · PhD Candidate

Computer-vision researcher and AI engineer based in Auckland. **Full-time AI Developer at TOHU MEDIA**, a Māori-owned media-production company, where I ship object-detection and OCR systems for film-industry video footage. Concurrently completing my PhD at **Auckland University of Technology** on graph neural networks for skeleton-based human activity recognition — first-author paper accepted at **Pattern Recognition** (Q1, 2026). Six peer-reviewed papers, 70+ citations on Google Scholar, four years of industry experience earned alongside postgraduate study, and eight years of cumulative classroom teaching. Looking for a senior **Computer Vision / Machine Learning Engineer** role with a New Zealand team building practical AI products.

TECHNICAL SKILLS

Programming
Python (expert) · Java

Deep Learning
PyTorch (primary) · TensorFlow / Keras · OpenCV · scikit-learn · NumPy · pandas · Hugging Face

Computer Vision
Object detection (incl. YOLO family) · object tracking · action recognition · OCR · semantic / instance segmentation · classification · video analytics

Research methods
Graph convolutional networks · attention & Transformers · skeleton-based human activity recognition · frequency-domain learning

Generative AI & LLM
LLM integration · agentic workflows · agent skills & tool use · Claude / GPT APIs · prompt engineering

Deployment
Docker · containerised model inference on cloud-hosted GPU servers

Tooling
Git / GitHub · Linux · CUDA · Jupyter

EDUCATION

PhD, Computer & Information Sciences
Auckland University of Technology
Oct 2023 – Oct 2026 (expected)
Thesis on deep learning for skeleton-based human activity recognition. **AUT PhD Full Scholarship** (Oct 2023 – Oct 2026).

Master of Computer & Information Sciences
Auckland University of Technology
2021 – 2023 · Specialisation: Computer Vision

BEng, Logistics Engineering (Software)
Dalian Neusoft University of Information, China
2011 – 2015

AWARDS & HONOURS

- **AUT PhD Full Scholarship** — tuition + stipend, Oct 2023 – Oct 2026
- SECMS Research Assistant Fund · 2023
- AUT Summer Research Assistant Scholarship · 2022

PEER REVIEW

- **ACM TOMM** — Multimedia Computing · 2024
- **International Agrophysics** · 2025, 2026
- **IEEE PIMRC** — Mobile Radio Comm. · 2026

SELECTED EXPERIENCE

Artificial Intelligence Developer *at TOHU MEDIA* Auckland · Feb 2022 – present

Full-time since June 2025; previously part-time alongside MSc and PhD

- AI engineer at a Māori-owned media-production company, responsible for the company's AI capability end-to-end — design and ship **object-detection and OCR systems** that operate on client video footage inside internal production pipelines (specific deliverables under NDA).
- Engineer for the **messy edge cases of real footage** that off-the-shelf CV models stumble on — variable cinematography and framing, dense overlay text in non-standard layouts, archival material with degraded contrast, and long-form takes where naive per-frame inference doesn't fit a single-workstation GPU budget. Pick model architecture, training regime, and post-processing accordingly.
- Own the **full ML lifecycle**: dataset scope and labelling spec, model training and evaluation in PyTorch, **Docker-packaged inference deployed to the company's cloud GPU servers**, and iteration with editors and the creative director on what "useful output" looks like inside an actual editing workflow.
- Make independent calls on architecture, **latency / accuracy trade-offs**, and scope reduction under deadline — across four years part-time alongside the MSc and PhD, and now full-time ownership of the AI work.

PhD Researcher *at Institute of Robotics & Vision (IoRV), AUT* Oct 2023 – Oct 2026

Supervisors: Dr Raymond Lutui (primary) · Assoc. Prof. Sira Yongchareon

- Designed **FATL-GCN**, a frequency-aware spatio-temporal graph network for skeleton-based human activity recognition; outperforms prior SOTA on NTU RGB+D 60 / 120 — accepted at *Pattern Recognition* (Q1 journal, IF = 7.6) in 2026.
- Authored **Dynamic Self-Attention Gated ST-GCN** at ICONIP 2024; introduces learned gating between spatial and temporal streams.
- **Full ownership of first-author research** — from idea, to model design, to GPU training, ablations, paper writing and rebuttal.
- Mentor master's students on PyTorch, experimental hygiene, and academic writing.

Research Assistant *at School of Engineering, Computer & Mathematical Sciences, AUT* Nov 2022 – Jul 2023

- Funded RA on the **Kiwifruit detection & counting** research line — co-built KiwiDetector (improved YOLOv7) and KiwiTracker for real-time orchard video; the lead paper now has **50+ citations**.
- Co-authored 3 peer-reviewed papers (IVCNZ 2022, SAI 2023, PSIVT 2023) during the assistantship.

Teaching Assistant *at AUT* 2022 – present

- TA on multiple undergraduate computer-science courses (programming, computer vision, AI). Run labs, mark assignments, hold consultation hours, and mentor on final-year projects.

Tech Service Specialist *at PB Tech* Auckland · Apr – Dec 2023

- Diagnosed hardware and software issues and supported walk-in customers in a busy retail tech-service environment; sharpened plain-English technical communication with non-specialist users.

Co-Founder · Junior Lecturer *at Baiyin Pinuo Training School · Baiyin Vocational & Technology College* China · 2015 – 2021

- **Junior Lecturer (2015 – 2019):** taught Computer Fundamentals, Java, IoT applications and RFID technologies — four years of full-classroom teaching to ~120 students per year.
- **Co-Founder (2019 – 2021):** co-founded a private training school; built curriculum, ran operations and finance; divested in 2021 to relocate to New Zealand for postgraduate study — the school remains in operation under my co-founder.

SELECTED PUBLICATIONS

6 peer-reviewed papers · 70+ citations · full list on ***Google Scholar***

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). “Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition.” *Pattern Recognition*. Q1 journal
doi.org/10.1016/j.patcog.2026.113146

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2025). “Dynamic Self-Attention Gated Spatial-Temporal GCN for Skeleton-Based HAR.” *ICONIP 2024*, Springer LNCS.
doi.org/10.1007/978-981-96-6588-4_14

Y. Xia, M. Nguyen, W. Q. Yan (2025). “An Improved YOLO Model for Kiwifruit Detection.” *Optimization, Machine Learning, and Fuzzy Logic*, IGI Global book chapter.
doi.org/10.4018/979-8-3693-7352-1.ch008

Y. Xia, M. Nguyen, W. Q. Yan (2022). “A Real-Time Kiwifruit Detection Based on Improved YOLOv7.” *IVCNZ 2022*, Springer LNCS. 50 citations
doi.org/10.1007/978-3-031-25825-1_4

Y. Xia, M. Nguyen, W. Q. Yan (2023). “Kiwifruit Counting Using KiwiDetector and KiwiTracker.” *SAI Intelligent Systems Conference*, Springer.
doi.org/10.1007/978-3-031-47724-9_41

Y. Xia, M. Nguyen, R. Lutui, W. Q. Yan (2023). “Multiscale Kiwifruit Detection from Digital Images.” *PSIVT 2023*.
doi.org/10.1007/978-981-97-0376-0_7

REFERENCES

Available on request. Doctoral supervisors (Dr Raymond Lutui, AUT; Assoc. Prof. Sira Yongchareon, AUT) and an industry referee from TOHU MEDIA can be provided.